Lost in Speech: Trilingual Spoken Hallucination Detection Across Audio and Transcripts

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Abstract

While text-based hallucination detection is well studied, reference-free detection of factual alterations in speech remains underexplored, especially for low-resource languages. Our spoken benchmark comprises 12,013 English, Russian, and Kazakh news samples with three synthetic alteration types and three severity levels, pairing source articles with rewrites as text, synthesized audio, and ASR transcripts. We add 290 fact-checked misinformation items collected in Russian (225) and Kazakh (65), translated into the other language and rendered through the same TTS–ASR pipeline. We evaluate fine-tuned multilingual encoders and zero-shot multimodal decoders on text, transcripts, and audio. Detectors receive only target inputs without source articles or external evidence; the task evaluates reference-free classification rather than evidence-grounded verification. Encoder degradation from source text to transcripts generally tracks per-language ASR error on the binary task. Among decoders, only Gemma-3n exceeds the binary majority-class baseline in macro-F1, on transcripts only; the other four do not exceed their respective baselines. Comparisons between audio and transcripts are confounded by differences in evaluation coverage and class balance. Synthetic-trained detectors achieve 0.82–0.88 macro-F1 on real-world misinformation source text; Russian provenance analysis reveals veracity-related and model-dependent machine-style signals, a key confound in synthetic hallucination benchmarks.

1 Introduction

In recent years, large neural models for language and speech have demonstrated impressive generative abilities, yet they exhibit a troubling propensity to hallucinate, producing information that is not grounded in factual sources (Frieske and Shi, 2024). In the spoken setting the stakes are higher still: transcription systems can emit entirely fabricated phrases, a non-trivial fraction of which carry explicit real-world harms (Koenecke et al., 2024). Despite this, hallucination detection has been studied almost exclusively in text (Ji et al., 2023), while work in the spoken domain remains scarce—and scarcer still for low-resource languages.

Unlike textual inputs, spoken content is processed through multi-stage pipelines that couple text-to-speech (TTS) synthesis and automatic speech recognition (ASR), introducing noise and cascading error propagation. The robustness of hallucination detection under realistic speech-pipeline conditions is therefore still insufficiently understood (Atwany et al., 2025). As large language and audio-language models are deployed ever more widely, evaluating hallucination detection under multilingual speech constraints has become increasingly critical (Frieske and Shi, 2024; Latif et al., 2023).

Three gaps remain open. First, dedicated benchmarks for hallucination in audio and audio-language models have only begun to emerge, and they are overwhelmingly English and question-answering oriented (Kuan et al., 2024; Cheng et al., 2025). Second, the few multilingual hallucination benchmarks are text-only (Abdaljalil et al., 2025; Vázquez et al., 2025), and recent evidence indicates that hallucination behavior does not track a language's digital footprint in any simple way (Obaid ul Islam et al., 2025)—motivating the study of a genuinely low-resource language such as Kazakh alongside higher-resource English and Russian. Third, existing hallucination benchmarks—synthetic by construction—have not been validated against real-world misinformation: whether detectors trained on LLM-generated hallucinations transfer to human-written false content remains untested, a question sharpened by the fact that in synthetic benchmarks text provenance (human vs. LLM) is perfectly correlated with the label. A further bottleneck is the lack of large, balanced, and annotated audio datasets, which is acute precisely where both

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TTS and ASR are least mature (Chun et al., 2025). To our knowledge, no prior resource combines spoken input, a *detection* task, three languages including Kazakh, and *controlled* hallucination types at graded severity; existing work covers only subsets of these axes.

For benchmark construction, we operationally define a *spoken hallucination* as a synthetically introduced factual or contextual alteration in a news article that is unsupported by, or contradictory to, its source article. The source article is used to generate and assign the benchmark labels but is not provided to the evaluated detectors. Thus, our experiments concern reference-free classification of benchmark-defined, source-relative alterations rather than evidence-grounded verification of factual truth. To study this problem under controlled yet realistic conditions, we construct a multilingual benchmark[1] spanning English (high-resource), Russian (medium-resource), and Kazakh (low-resource). Each sample pairs a reference article with a systematically generated hallucinated counterpart of controlled hallucination type (fabrication, contradiction, context inconsistency) and severity level (mild, moderate, severe), while preserving topic and style. Our controlled generation and LLM-as-a-judge validation descend from synthetic-hallucination pipelines developed for text (Mishra et al., 2024; Xie et al., 2024). To simulate deployment, every version is provided in both text and speech form: hallucinated texts are synthesized with language-specific TTS systems and transcribed back with ASR (Fig. 1). This design enables evaluation on clean text, ASR-transcribed text, and raw audio, and supports analysis of how cascading speech-pipeline noise—a documented risk in spoken language understanding (Avila et al., 2023)—affects reference-free classification of the benchmark labels. We evaluate two model families on the resulting data: fine-tuned multilingual encoders and zero-shot multimodal (audio-language) decoders. Our experiments show TTS–ASR pipeline degradation most pronounced for Kazakh.

Contributions.

1. A multilingual spoken benchmark for reference-free classification of synthetically introduced factual and contextual alterations, with controlled alteration types and graded severity: 12,013 news samples comprising source articles and aligned rewritten counterparts in English, Russian, and Kazakh, represented as text, synthesized speech, and ASR transcripts.
2. A structured LLM-based hallucination generation framework with generator self-assessment and external validation by two independent cross-family LLM judges, supplemented by targeted human inspection of synthesized speech.
3. The first real-world spoken evaluation split for this task: 290 fact-checked fake news stories from factcheck.kz, collected natively in Russian (225) and Kazakh (65), each paired with a translation into the other language and with matched truthful negatives, rendered through the same TTS→ASR pipeline (Section 3.2).
4. A Russian provenance analysis that probes factuality and human-vs-machine text signals—a confound inherent to synthetic benchmarks—by evaluating detectors on human-written false content (Section 4.3).
5. A language-specific TTS→ASR pipeline with best-of-breed per-language ASR, and a comprehensive empirical study of fine-tuned encoders and zero-shot audio-language decoders across text and audio settings, with ASR error analysis and an analysis of decoder failure modes, output parseability, and the input-length limits of audio encoders, addressing RQ1–RQ3 (Section 4).

2 Related Work

Hallucination detection in text and speech. Text-based hallucination research spans error taxonomies (Ji et al., 2023; Huang et al., 2025), synthetic supervision (Mishra et al., 2024; Xie et al., 2024), and reference-free detection through sampling consistency (Manakul et al., 2023). Speech studies examine model-generated errors (Frieske and Shi, 2024; Atwany et al., 2025) and associated harms (Koenecke et al., 2024), while audio benchmarks assess object hallucination and audio question answering (Kuan et al., 2024; Cheng et al., 2025). Multilingual hallucination benchmarks (Abdaljalil et al., 2025; Vázquez et al., 2025; Obaid ul Islam et al., 2025) focus on text.

[1] https://huggingface.co/datasets/maristombayeva/lost-in-speech

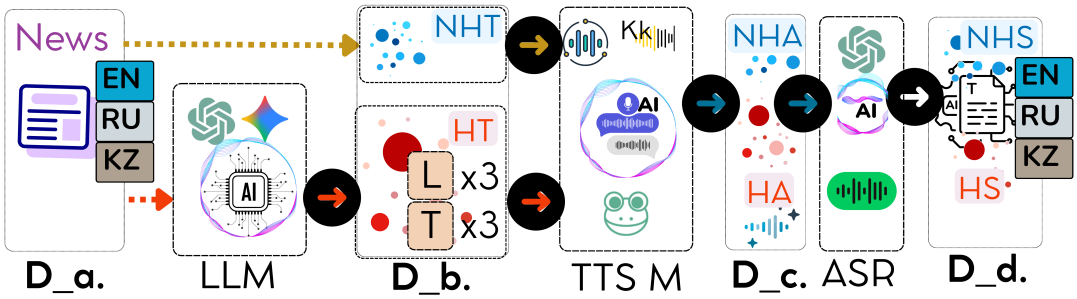


Figure 1: Data generation pipeline framework. **Legend:** D_a = Original Data, D_b = Generated Hallucinated Text, D_c = Audio Data, D_d = Transcribed Data; T = Type, L = Severity Level; NHT = Non-Hallucinated Text, HT = Hallucinated Text, NHA = Non-Hallucinated Audio, HA = Hallucinated Audio, NHS = Non-Hallucinated Speech, HS = Hallucinated Speech; LLM = Large Language Model, TTS M = Text-to-Speech Model, ASR = Automatic Speech Recognition; MT = Machine Translation.

Fact-checking and spoof detection. FEVER (Thorne et al., 2018), AVeriTeC (Schlichtkrull et al., 2023), and X-Fact (Gupta and Srikumar, 2021) support evidence-grounded claim verification. Our detectors predict source-relative alteration labels without access to source articles or external evidence. FakeSV (Qi et al., 2023) and MAD (Chun et al., 2025) address video misinformation and spoken dialogue, respectively. Spoof detection (Wang et al., 2024a; Yi et al., 2023; Zhang et al., 2023) targets speech authenticity rather than factual or contextual alterations.

Positioning. Audio-language models (Chu et al., 2024) and ASR systems (Radford et al., 2023) enable direct audio and transcript-based processing; AIR-Bench (Yang et al., 2024) evaluates audio comprehension and instruction-following. Kazakh resources support recognition and synthesis (Khassanov et al., 2021; Mussakhojayeva et al., 2021, 2022) and knowledge evaluation (Togmanov et al., 2025). Our benchmark combines English, Russian, and Kazakh text, synthesized speech, and ASR transcripts for reference-free classification of controlled factual and contextual alterations by type and severity. We also test real-world transfer and examine text provenance—the target of machine-generated text detection (Wang et al., 2024b)—as a potential confound.

3 Benchmark Construction

3.1 Synthetic Subset

We constructed a corpus of 12,013 samples based on news articles gathered from major Kazakhstani news platforms (e.g., nur.kz, forbes.kz, tengrinews.kz, kapital.kz), covering diverse topics (politics, economy, technology); each sample is either an original article or one of its generated hallucinated versions (Table 1). The articles span three languages—Kazakh, Russian, and English—and are treated as factual references since they originate from established outlets. We generated hallucinated versions with three hallucination types (fabrication, contradiction, and context inconsistency) and three severity levels (mild, moderate, severe), following Huang et al. (2025): the English batches were produced with `gpt-3.5-turbo` and `gpt-4`, and the Russian and Kazakh batches with `gemini-2.0-flash-lite`; the prompts of the final pipeline, including all language variants, are documented in Appendix A. Table 1 summarizes the dataset distribution. Each sample is available as text, synthesized audio, and ASR transcript (Fig. 1). As speech is produced via a TTS→ASR pipeline, transcription errors reflect cascading noise from both synthesis and recognition components, particularly in low-resource Kazakh.

To create diverse and controlled hallucinations, we implemented a category-based prompting framework inspired by factuality taxonomies in prior work (Huang et al., 2025; Lucas et al., 2023; Nahar et al., 2024). Each article was rewritten according to a predefined hallucination type and severity level.

We distinguish *Factual Contradiction* (statements directly conflicting with facts or information from the original article), *Factual Fabrication* (in-

Table 1: Dataset summary across languages. Res = resource level (H = high, M = medium, L = low). Type: C = contradiction, F = fabrication, I = inconsistency. Severity: Mild, Moderate, Severe. Rows marked "real" denote the real-world evaluation split from factcheck.kz (Section 3.2); †each language combines natively collected fakes (225 ru, 65 kz) with parallel translations of the other language's items. Real items carry binary labels only and are used exclusively for evaluation. The 290 truthful items of each real row are original articles of the synthetic subset above (not additional data); only the 290 fakes per language are new items.

Lang	Res	Total	Binary		Type			Severity		
			No	Yes	C	F	I	Mild	Mod	Sev
en	H	3967	1340	2627	903	938	786	875	877	875
kz	L	3978	1113	2865	947	956	962	957	953	955
ru	M	4068	1259	2809	962	960	887	934	937	938
ru (real)†	M	580	290	290	–	–	–	–	–	–
kz (real)†	L	580	290	290	–	–	–	–	–	–

sertion of fabricated yet plausible-sounding details not grounded in the source), and *Context Inconsistency* (subtle alterations that distort the meaning, emphasis, or context of the original article without explicit factual errors).

Each generation was guided by a structured prompt incorporating the original article, the target hallucination type, and severity level. The generation prompt also required the generating model to provide a structured self-assessment of its output along predefined quality dimensions. Because such in-prompt self-assessment is not independent of generation, we additionally conducted a separate external validation stage using two judge models from families disjoint from the generators (Section 3.4). All generation and evaluation prompts are provided in Appendix A.

3.2 Real-World Subset

Synthetic benchmarks carry an inherent confound: every hallucinated sample is LLM-generated while every faithful sample is human-written, so provenance is perfectly correlated with the label. To reduce this direct human-versus-LLM label correlation and to test performance on naturally occurring misinformation, we construct a real-world evaluation split from factcheck.kz, a professional Kazakhstani fact-checking organization.

We collected 290 news items verified as false by factcheck.kz, downloaded from its public archive in October 2024: 225 published natively in Russian and 65 natively in Kazakh (after removing 15 exact duplicates from the initial Russian collection), spanning topics comparable to the synthetic corpus (politics, society, health, technology). Each item consists of the fake claim or article as it circulated, with the fact-checking verdict as ground truth. To enable controlled cross-lingual comparison on identical content, every item is additionally translated into the other language using Google Translate, yielding a parallel Russian–Kazakh fake-news corpus of 290 stories per language, of which 65 Kazakh and 225 Russian items are natively circulating misinformation. Six claims were independently fact-checked in both languages; these near-parallel native pairs are flagged in the release. Translation quality was manually validated by a fluent bilingual author on all 290 translations, all of which were judged adequate (Appendix E).

As truthful negatives, we select 290 human-written original articles per language from the synthetic subset, matched to the fakes via greedy TF-IDF cosine similarity over topic and time period. These articles are excluded from the training and development splits of every model evaluated on the real-world data (Appendix B), so no model evaluated on this split has seen any of its items during training. All real-world items—fake and truthful—are rendered through the identical TTS→ASR pipeline described in Section 3.3, so the real-world split was evaluated in the same three modalities as the synthetic corpus (only URLs, labels, and metadata are released; see Ethical Considerations). This split is used exclusively for evaluation; no model is trained on it. Unlike the synthetic subset, real-world items carry only binary labels, since fact-checkers do not annotate our type/severity taxonomy; manual taxonomy alignment is left to future work.

3.3 Speech Synthesis and Transcription

We synthesized speech for both original and hallucinated texts using language-specific TTS systems: Coqui XTTS-v2 (Coqui AI, 2023) for English, Silero TTS (Silero Team, 2021) for Russian, and an open-source Kazakh TTS model (Mussakhojayeva et al., 2021). Multiple voices were employed to increase acoustic diversity. Due to limited availability of high-quality Kazakh TTS systems, synthesized Kazakh speech exhibited reduced naturalness and occasional phonetic instability. Hallucination labels are assigned at the text-generation stage relative to the source article; subsequent TTS–ASR discrepancies are treated as pipeline degradation and do not alter these labels.

All audio samples were transcribed using ASR.

English and Russian were processed using Whisper-large-v3, while Kazakh was transcribed using a fine-tuned wav2vec2-large model with post-processing for punctuation. We adopt the strongest available ASR per language: as shown in Table 7 (Appendix C), Whisper-large-v3 is highly accurate on English and Russian but fails on low-resource Kazakh, whereas the fine-tuned wav2vec2 model nearly halves the Kazakh error rate on identical audio. All WER/CER values are computed at the corpus level after lowercasing, punctuation removal, and digit removal, applied identically across languages.

We quantify transcription distortion using WER/CER between source texts and ASR outputs: EN 7.30%/2.74%, RU 9.08%/4.74%, and KZ 34.04%/16.90% (overall WER 13.02%, CER 6.46%). The substantially higher Kazakh error rates reflect compounded degradation across the TTS–ASR cascade. In particular, phonetic distortions introduced by TTS in named entities and region-specific terms (e.g., Oskemen realized as Askemin) are faithfully transcribed by ASR but counted as full word errors against the original text. The news domain further amplifies this effect due to a high density of proper names and Kazakhstan-specific terminology (e.g., akimat), which are underrepresented in multilingual training corpora. Additionally, Kazakh's agglutinative morphology increases lexical variability, so even minor deviations often result in complete word-level mismatches, inflating WER relative to CER.

This best-of-breed choice minimizes, but does not eliminate, the cross-language tooling gap; the residual gap (Kazakh WER remains roughly four times higher than English and Russian) is examined by relating per-language ASR error to downstream detection degradation (Sections 4.1 and 4.2).

3.4 Quality Evaluation: LLM vs. Human

To validate the quality of both raw hallucinated texts and synthesized speech, we perform two tasks.

Evaluating Hallucinated Texts. We adopted an LLM-as-a-Judge framework (Zheng et al., 2023; Li et al., 2024; Gu et al., 2025). Generation prompts included an in-prompt self-assessment; independent validation used a stratified sample of 216 articles ($n = 8$ per type$\times$severity$\times$language cell) and two external judges, Claude Haiku 4.5 and DeepSeek Chat, under the same protocol.

The evaluation considered nine dimensions, applied type-dependently (seven per type): Adequacy of Type (AT), Fluency (Fl), Coherence (Co), Logical Coherence (LC), Internal Consistency (IC), Credibility (Cr), Factual Accuracy (FA), Relevance (Re), and Satisfaction (Sa); definitions are in Appendix A.

Each criterion was rated on a five-point ordinal scale from *Very Low (1)* to *Very High (5)*. Table 10 (Appendix D) reports mean Claude scores on this sample by type, severity, and language. As the scale is ordinal, means serve only as descriptive indicators.

For both factuality-bearing types, the judge assigns low credibility and factual accuracy already at mild severity ($\sim$1.2–2.0) and floor scores ($\approx$1.0) at severe levels, confirming that the intended factual degradation is realized. Adequacy of Type and Satisfaction increase with severity for all three types, reaching 4.9–5.0 at the severe level, so generated articles match their target category and level most clearly where distortion is strongest. Fluency declines only moderately with severity and does not collapse for any type (severe-level means of 2.6–4.0 for contradictions; Table 10), indicating that the framework manipulates semantic grounding rather than linguistic form.

Cross-Judge Validation. A natural concern is that the quality scores in Table 10 could reflect judge-specific idiosyncrasies. To test this, we measured agreement between the two independent external judges, Claude Haiku 4.5 and DeepSeek Chat, on the same stratified sample and per quality dimension (Table 9, Appendix D). Agreement is high on the two factuality-oriented dimensions that underlie the hallucination signal—Factual Accuracy and Credibility (quadratic-weighted $\kappa = 0.875$ and 0.897; Spearman $\rho = 0.91$ and 0.92)—with the two judges agreeing within one point on 100% of these items. For the remaining, more subjective dimensions, raw agreement remains high (within one point on 71–97% of items), although weighted κ is lower for the most concentrated scales (e.g., Internal Consistency, Logical Coherence), where limited score variance deflates chance-corrected agreement. Overall, the cross-judge results corroborate the trends in Table 10, indicating that the reported quality patterns are robust across the two external evaluators.

Evaluating Synthesized Speech. Given Kazakh's high WER/CER, a fluent native Kazakh-

speaking author randomly selected and evaluated 120 audio samples using predefined criteria for intelligibility, faithfulness to the source text, and exclusion due to audio quality.

The results indicate that 94.2% of the samples were rated as mostly intelligible and 5.8% as poorly intelligible, with no samples classified as fully unintelligible. Regarding semantic fidelity, 61.7% exhibited minor discrepancies and 38.3% major discrepancies relative to the source text. No samples were excluded due to critical audio quality issues. These findings confirm that the Kazakh pipeline introduces noticeable semantic distortion while maintaining general perceptual comprehensibility, thereby motivating focused robustness analysis in the low-resource setting.

4 Empirical Evaluation

We aim to answer the following research questions: *RQ1: How robust are state-of-the-art multilingual encoder models for reference-free classification of synthetically introduced factual alterations when evaluated on original text versus ASR-transcribed text? RQ2: Under identical zero-shot prompting conditions, can multimodal decoder models perform reference-free classification from ASR transcripts and from direct audio, and are the two inputs comparable on this benchmark? RQ3: Do classifiers trained on synthetically altered texts transfer to real-world, human-written misinformation, and to what extent is their binary signal attributable to human-vs-machine text discrimination rather than veracity-related content?*

To address RQ1, a model receives only the target news item, represented either as text (D_a) or as an ASR transcript (D_d), and predicts labels assigned during controlled dataset construction. The corresponding source article and external evidence are not provided at inference time. We evaluate binary classification (unaltered vs. synthetically altered) and multiclass classification of the assigned alteration type and severity. Consequently, these experiments measure reference-free recognition of benchmark-defined alterations and should not be interpreted as evidence-grounded factual verification. We compare (i) fine-tuned multilingual encoder-based transformers and (ii) zero-shot in-context learning with decoder-based architectures. The models and modality support are summarized in Table 8 (Appendix C).

Throughout, we report accuracy and macro-F1; training details are given in Appendix B.

Table 2: Finetuned encoder performance (Acc/F1). Task: B = Binary, T = Type, L = Level; Split: O = original text, T = ASR-transcribed text. Bold indicates the best F1 within each row (among the four encoders).

Task	Lan	Split	XLM-R (b)	XLM-R (l)	mDeBERTa	ReMBERT
			Acc/F1			
B	en	O	.826/.828	**.865/.863**	.843/.843	.879/.860
		T	.838/.832	.658/.522	**.840/.841**	.834/.815
	kz	O	.870/.873	.841/.846	**.893/.893**	.910/.876
		T	.826/.826	.711/.591	**.875/.874**	.802/.782
	ru	O	.809/.814	.738/.747	.838/.841	**.862/.845**
		T	**.832/.835**	.703/.580	.824/.827	.788/.767
T	en	O	.612/.599	**.680/.679**	.600/.596	.388/.187
		T	.571/.564	**.673/.666**	.627/.623	.331/.166
	kz	O	.612/.606	.675/.674	**.681/.676**	.306/.156
		T	.260/.138	**.635/.636**	.559/.560	.347/.172
	ru	O	.669/.663	**.665/.665**	.616/.616	.345/.171
		T	**.676/.677**	.665/.660	.645/.644	.353/.174
L	en	O	.596/.590	**.616/.611**	.604/.596	.344/.171
		T	**.633/.616**	.290/.150	.568/.570	.394/.188
	kz	O	.592/.592	**.660/.660**	.664/.656	.326/.163
		T	.559/.555	.374/.182	**.601/.586**	.297/.153
	ru	O	.624/.624	**.668/.655**	.598/.595	.336/.168
		T	.618/.611	.333/.167	**.633/.634**	.316/.160

4.1 RQ1 → Text-based: Original Text vs. Transcripts

We assess multilingual encoder robustness in text-only deployment scenarios. Specifically, we compare performance on original articles (D_a) and their ASR-transcribed counterparts (D_d).

Table 2 shows that fine-tuned encoders achieve strong binary hallucination detection performance (F1: 0.52–0.89), while fine-grained type (F1: 0.14–0.68) and severity classification (F1: 0.15–0.66) remain more challenging. Performance is predominantly higher on original text than on ASR-transcribed content, though not uniformly so: in a few language–task cells transcripts match or exceed original text (e.g., XLM-R (base) on Russian). mDeBERTa exhibits the most robust cross-lingual performance; ReMBERT is competitive on binary detection but performs poorly on type and severity tasks (F1 .153–.188). For mDeBERTa and ReMBERT, binary-task degradation from D_a to D_d increases with per-language ASR error, whereas this trend is inconsistent for type and severity tasks and XLM-R models.

4.2 RQ2 → Audio-Based: Decoders on Audio and Transcripts

We evaluate five decoder models (1.5B–8B) in a zero-shot setting, reporting ASR-transcript (D_d) results for all five and direct-audio (D_c) results for

three (Table 3). We report direct prompting only. Under chain-of-thought (CoT) prompting, Qwen2-Audio requested audio in every fine-grained audio call, Qwen2.5-Omni refused 33% of Kazakh binary audio calls (versus 2% under direct prompting), and up to 78% of Gemma-3n's responses were unparsable. Coverage differs across cells (see caption), limiting direct comparisons.

Near-chance scores reflect degenerate output rather than partial competence, and the mechanism differs by model. Qwen2.5-Omni parses over 96% of transcript calls but rarely asserts an alteration: on Kazakh its binary predictions are constant negative—accuracy .279 on the full corpus, matching the unaltered share up to unparsed items, and .834 on the 344-item audio subset, where the unaltered share is .854 and the remaining 2% are refusals counted as errors. Qwen2-Audio and Step-Audio-2-mini are also constant on the balanced binary subsets (.500/.333 in all languages), though Qwen2-Audio exceeds the fine-grained macro-F1 baseline on transcripts (.198–.357) and Step-Audio-2-mini on English type (.467/.401). LFM2-Audio, documented for English only, fails at the output level. Gemma-3n is the only decoder above the binary baseline on transcripts in all three languages, with macro-F1 .613/.581/.630 against .40–.42, and it also exceeds the severity baseline (.399/.328/.370 against .167); its type predictions stay near baseline.

We do not compare modalities. Audio items have a median duration of 96 s (English: 212 s; 98% exceed 30 s). Qwen2-Audio's Whisper-large-v3 front end is documented to truncate beyond 30 s; Qwen2.5-Omni uses a block-wise streaming encoder and LFM2-Audio a FastConformer encoder, neither documented with that limit. We did not verify per-model truncation in our inference code, so under direct audio a model may receive only part of an article labelled over its full text. The audio subsets are also skewed—the Kazakh one is 85% unaltered against 28% in the corpus—so their accuracies are not comparable to corpus baselines. For Kazakh, 38% of inspected audio also shows major semantic discrepancies from the source text (Section 3.4), so the text-level label may not be recoverable from audio at all. The near-constant audio predictions are consistent with these mismatches and say nothing about audio reasoning.

LFM2-Audio and Step-Audio-2-mini are evaluated outside their documented language coverage on Russian and Kazakh by design, to show where open audio-language models stop working.

4.3 RQ3 → Real-World Transfer and the Provenance Confound

Real-world transfer. We take the strongest fine-tuned encoders from Section 4.1 (ReMBERT and mDeBERTa), trained exclusively on the synthetic subset, and evaluate them on the real-world split (Table 4). On original text, binary macro-F1 on real-world data matches or exceeds synthetic test performance for both encoders and both languages (e.g., ReMBERT: 0.819 vs. 0.786 for Russian, 0.883 vs. 0.863 for Kazakh), showing that the synthetic-trained detectors retain high classification performance on this real-world split. The same pattern holds on ASR transcripts: with encoders fine-tuned on transcribed text, real-world macro-F1 remains high (.753–.835), and the O→T degradation grows with per-language ASR error ($\leq$6 points for Russian—absent for ReMBERT, whose transcript fine-tuning even compensates for ASR noise—and up to 12.5 points for Kazakh), mirroring the RQ1 trend. On the 290 parallel Russian–Kazakh fake pairs, mDeBERTa's predictions disagree on only 3.8% of items, indicating substantial cross-lingual stability on identical false content.

Provenance-controlled analysis. We conduct this analysis on Russian original text. In the synthetic binary task, provenance (human vs. LLM) is perfectly correlated with the label; a detector could therefore succeed by recognizing machine-generated style alone. We complete the provenance×veracity matrix (Table 5) with the real-world fakes (human-written false) and with 290 *faithful* LLM rewrites of the same truthful articles, generated with Gemini 2.0 Flash-Lite under a hallucination-free instruction. The results reveal evidence of two components. First, a veracity component: within human-written text, both encoders separate false from truthful articles by a wide margin (flag rates 1.000 vs. 0.352 for ReMBERT; 0.962 vs. 0.221 for mDeBERTa). Second, a provenance component whose strength is model-dependent: rewriting a truthful article—changing style while preserving every fact—raises ReMBERT's flag rate from 0.352 to 1.000 (saturation on machine-generated text), but mDeBERTa's only to 0.586, which remains well below its 0.893 rate on synthetic hallucinations—i.e., mDeBERTa retains veracity sensitivity even within machine-generated text. Synthetic binary F1 therefore overstates fac-

Table 3: Decoder performance on direct audio (A) and ASR transcripts (T). Direct prompting only; chain-of-thought output was unusable (Section 4.2). Acc and macro-F1 use the same items; unparsable and *none* responses count as errors. Cells with different n are not comparable. Transcript n: full corpus for •, ▲, ■ (3,965/3,977/4,067 binary; 2,627/2,865/2,809 fine-grained, en/kz/ru); 100/150 for ⧫; 60/90 for ⋆. Audio n: full corpus for • on en, 344/150 kz, 100/150 ru; 100/150 for ▲; 60/90 for ⧫; ⋆ transcripts only, ■ audio excluded (raw responses not retained) (–). Majority baselines (Acc/macro-F1): .663/.399, .720/.419, .691/.409 (full binary corpus, en/kz/ru); .500/.333 balanced binary subsets; full-corpus type .357/.176, .336/.168, .343/.170 and severity .334/.167 (en/kz/ru); .333/.167 on the balanced fine-grained subsets. **Bold**: best among the full-corpus models (•, ▲, ■) on transcripts, the only comparable cells. †audio not delivered in 7–13% of calls; not interpretable. M: ⧫ = LFM2-Audio-1.5B, ▲ = Qwen2-Audio-7B-Instruct, • = Qwen2.5-Omni-3B, ⋆ = Step-Audio-2-mini, ■ = Gemma-3n-E4B-it.

		Binary		**Type**		**Level**	
M	**Lang**	**A**	**T**	**A**	**T**	**A**	**T**
		(Acc / F1)					
⧫	en	.250/.336	.140/.186	.000/.000	.013/.024	.000/.000	.080/.114
	kz	.183/.254†	.190/.254	.000/.000	.013/.024	.000/.000	.000/.000
	ru	.067/.094†	.170/.207	.011/.020	.013/.025	.000/.000	.007/.013
▲	en	.500/.333	.391/.339	.333/.167	**.313/.207**	.000/.000	.423/.357
	kz	.500/.333	.294/.250	.333/.167	**.328/.210**	.000/.000	.327/.280
	ru	.500/.333	.346/.296	.333/.167	**.286/.198**	.000/.000	.280/.289
•	en	.364/.295	.359/.288	.032/.058	.024/.047	.106/.148	.146/.182
	kz	.834/.455	.279/.219	.000/.000	.030/.050	.013/.025	.021/.036
	ru	.510/.355	.339/.285	.013/.026	.038/.070	.053/.084	.088/.122
⋆	en	–	.500/.333	–	.467/.401	–	.400/.304
	kz	–	.500/.333	–	.333/.168	–	.333/.167
	ru	–	.500/.333	–	.333/.277	–	.278/.211
■	en	–	**.683/.613**	–	.311/.165	–	**.468/.399**
	kz	–	**.607/.581**	–	.262/.154	–	**.340/.328**
	ru	–	**.656/.630**	–	.256/.173	–	**.403/.370**

tuality sensitivity, the degree of provenance reliance is itself a model property, and the real-world split—where both classes originate from human-written content—substantially reduces the original provenance confound.

5 Insights and Challenges

1) Transcription quality is a key factor. Fine-tuned encoders generally degrade on ASR transcripts (D_d) relative to original text (D_a). For mDeBERTa and ReMBERT this binary-task degradation grows with per-language ASR error, largest in Kazakh, but the trend is less consistent for fine-grained tasks.

2) Zero-shot decoders mostly fail, but not uniformly.

Of five decoders (1.5B–8B), only Gemma-3n exceeds the majority-class baseline on the binary task in macro-F1, on transcripts (.613/.581/.630 vs. .40–.42; its audio responses were not retained). The other four do not exceed their respective binary macro-F1 baselines on transcripts, though Qwen2-Audio exceeds the fine-grained baselines (.198–.357 vs. .167–.176). Several models evaluated outside their documented language coverage produce near-constant outputs, so zero-shot multilingual audio reasoning is limited first by language coverage and output reliability, and only then by task granularity.

3) Low-resource conditions amplify pipeline challenges. For the fine-tuned encoders, Kazakh has the highest ASR error and pronounced downstream degradation, consistent with compounded TTS→ASR noise.

4) Granularity increases difficulty.

Binary classification is substantially easier than predicting alteration type and severity; the low fine-grained scores show that these construction-defined categories remain hard to distinguish reference-free.

5) Direct audio and transcripts are not comparable here. Evaluation coverage and class balance differ across audio and transcript conditions. Audio items are long (median 96 s), but we did not verify whether inputs were truncated during inference; any resulting information loss remains a possible, model-dependent limitation. We therefore report audio results without directly comparing them to

Table 4: Binary detection (macro-F1) on the synthetic test split vs. the real-world split, for original text (O) and ASR transcripts (T). Encoders are trained on the synthetic subset only, in a dedicated run whose train/dev pool excludes the 290 truthful negatives per language (Appendix B); the synthetic test split is augmented with these negatives, so the truthful class of the Synth (test) and Real rows shares the same 290 articles per language. O and T columns use encoders fine-tuned on original text and on ASR transcripts, respectively, over identical train/dev/test ids. These figures stem from an independent fine-tuning run and are not directly comparable to Table 2, which reports the full corpus split.

		ReMBERT		mDeBERTa	
Lang	**Split**	**O**	**T**	**O**	**T**
ru	Synth (test)	.786	.819	.825	.771
	Real	.819	.835	.870	.811
kz	Synth (test)	.863	.794	.831	.748
	Real	.883	.818	.878	.753

Table 5: Provenance×veracity analysis: fraction of items predicted "hallucinated" by ReMBERT (Russian, original text). Human/Truthful = truthful negatives of the real-world split (Section 3.2); Human/False = factcheck.kz fakes; LLM/Truthful = faithful Gemini 2.0 Flash-Lite rewrites of the same truthful articles; LLM/False = synthetic hallucinations (test split). mDeBERTa shows the same pattern with a weaker provenance component (human-written .221/.962; LLM-generated .586/.893).

	Truthful	**False**
Human-written	.352	1.000
LLM-generated	1.000	.986

transcript results.

6) The binary signal reflects both veracity and provenance. Detectors trained on synthetic hallucinations retain high performance on the real-world split, with real-world F1 matching or exceeding synthetic-test F1 for both encoders and languages. However, in the Russian original-text analysis, faithful LLM rewrites of truthful articles reveal a model-dependent provenance component: rewriting a truthful article without altering any fact raises ReMBERT's flag rate from .352 to 1.000, so its binary signal is in large part machine-text detection, whereas mDeBERTa's rises only from .221 to .586, well below its .893 on synthetic hallucinations. Synthetic binary scores thus overstate factuality sensitivity, and provenance-controlled analyses are needed to quantify this reliance, although other source-, register-, and distribution-related confounds may remain.

Overall, the benchmark studies reference-free recognition of controlled, text-level alterations rendered via TTS, not spontaneous speech or evidence-grounded verification. This controlled setup enables systematic analysis of speech-pipeline and resource effects on text-based detection, while exposing current limitations of zero-shot audio-language models. The real-world split partially relaxes this constraint by grounding evaluation in naturally occurring misinformation.

6 Conclusion

We introduce a multilingual spoken benchmark for reference-free classification of synthetically introduced factual and contextual alterations, comprising 12,013 aligned news samples in English, Russian, and Kazakh. The resource represents source and rewritten articles as text, synthesized speech, and ASR transcripts, and is complemented by a real-world evaluation split of fact-checked misinformation in Russian and Kazakh. It supports evaluation across text, transcript, and audio without claiming evidence-grounded verification of factual truth.

Results show that performance is strongly influenced by input representation and language resources: for fine-tuned encoders, original text generally outperforms ASR transcripts, with pronounced degradation in low-resource Kazakh. In zero-shot settings, only Gemma-3n beat the binary majority-class baseline (macro-F1) on transcripts (its audio responses were not retained); the other four decoders did not exceed their respective baselines. On the real-world split, synthetic-trained detectors retain high classification performance, while the Russian provenance analysis reveals both veracity-related and model-dependent machine-style signals.

We identify three main challenges: robustness to speech pipeline noise, amplified degradation in low-resource settings, and difficulty in fine-grained type and severity classification. Future work should focus on noise-robust multimodal modeling, provenance-robust detection objectives, and encoders that handle long-form audio.

Limitations

First, the benchmark primarily uses read speech synthesized with language-specific TTS systems rather than spontaneous or natively recorded speech. Its ASR transcripts consequently reflect errors introduced jointly by synthesis and recog-

nition, and the observed trends may not transfer directly to natural-speech deployment. Human validation is limited to 120 Kazakh samples and does not cover all languages, voices, or experimental conditions.

Second, WER is not directly comparable across typologically different languages because it is sensitive to tokenization, segmentation, and morphology. In agglutinative Kazakh, a minor morphological deviation can produce a full word-level error. We therefore report CER alongside WER and relate per-language ASR error to downstream performance rather than treating raw WER as a direct measure of language difficulty.

Third, the benchmark covers only English, Russian, and Kazakh within the news domain. Moreover, the English alterations were generated using GPT models, whereas the Russian and Kazakh alterations were generated using Gemini. Language is therefore partially confounded with generator identity, and cross-lingual differences cannot be attributed exclusively to language or resource availability. External validation by two cross-family LLM judges and targeted human inspection reduce, but do not eliminate, annotation, prompt, or generator-specific artifacts.

Fourth, provenance is correlated with the binary label in the synthetic subset: original articles are human-written, whereas altered articles are LLM-rewritten. A classifier may therefore respond partly to machine-generated style rather than exclusively to veracity-related content. We examine this confound in Section 4.3 using human-written misinformation and faithful LLM rewrites. The type and severity tasks do not share this specific binary confound, although they may retain prompt- or generator-specific cues.

Fifth, the evaluated models operate reference-free: they receive the target text, transcript, or audio, but neither the source article nor retrieved evidence. We also do not control whether individual news facts precede or follow each model's training-data cutoff. The results should therefore be interpreted as reference-free classification of synthetic alterations defined relative to source articles, rather than evidence-grounded, open-world factual verification.

Sixth, the real-world split contains 225 natively Russian and 65 natively Kazakh misinformation items, translated into the other language to produce 290 items per language. The larger translated share of the Kazakh split may introduce translation artifacts, and the split excludes English and is intended only for evaluation. In addition, fact-checked misinformation is not a random sample, and the real-world positives differ from the truthful negatives in source and register as well as veracity. High performance on this split may therefore partly reflect distributional cues. The provenance-controlled analysis reduces the direct human-versus-LLM confound but does not eliminate all such differences.

Finally, the decoder models are evaluated with zero-shot prompting and without task-specific audio fine-tuning. None of them documents support for Kazakh, and the two Qwen models do not report the languages their speech encoder covers, so results on these languages partly reflect model coverage rather than task difficulty. Their results characterize out-of-the-box performance under the evaluated conditions and should not be interpreted as an upper bound on the usefulness of audio. Fine-tuned or evidence-grounded multimodal models may exhibit different modality patterns.

Ethical Considerations

The benchmark is built from news articles published by public Kazakhstani outlets (nur.kz, forbes.kz, tengrinews.kz, kapital.kz). To respect third-party copyright, we do not redistribute the original article texts or their verbatim derivatives. For original articles and real-world fake items, the release contains only source URLs, labels, and metadata; ASR transcripts and synthesized audio are released only for LLM-rewritten items and can be regenerated for the original texts with the publicly available TTS and ASR systems described in Section 3.3. The release also includes our LLM-generated hallucinated rewrites, type/severity labels, and the split manifests.

For the small fraction of articles whose original pages are no longer available online, we provide the source outlet and publication metadata in place of a URL.

The original article texts are used internally solely as factual references. Because the rewrites necessarily retain part of the source content, their redistribution is subject to the limitations set out in Article 19 of the Republic of Kazakhstan Law on Copyright and Related Rights (No. 6-I), including attribution of each item to its source and use limited to the extent justified by the research purpose. The corpus consists only of already-published news and contains no private or sensitive personal data.

The real-world split is built from misinformation items publicly identified and debunked by factcheck.kz, a professional Kazakhstani fact-checking organization. We do not redistribute the fake texts, their translations, or their ASR transcripts and synthesized audio; the release contains only URLs of the published fact-checking reports, binary labels, and translation metadata, following the same scheme as for the original articles. Because every item in this split has already been publicly debunked and no fake text is redistributed, our resource does not introduce new misinformation into circulation.

Because the resource contains deliberately fabricated and contradictory content, it carries a potential for misuse (e.g., as disinformation). Rewrites may also attribute fabricated statements to real, named individuals and organizations; every rewritten item is therefore labeled as a synthetic alteration, and the dataset card states that rewrites must not be presented as factual reporting. The resource is intended solely for developing and evaluating detection systems and is distributed under a custom research-use license (`lost-in-speech-research-use-license`; full text in the `LICENSE` file of the dataset repository).

We used generative models (`gpt-3.5-turbo`, `gpt-4`, and `gemini-2.0-flash-lite`) to generate hallucinated text and `claude-haiku-4-5-20251001` and `deepseek-chat` as independent LLM-based judges for partial quality evaluation; this use is disclosed here in accordance with ACL policy on the use of generative AI.

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A Generation and Evaluation Prompts

This appendix documents, verbatim, the prompt templates of the final pipeline. Runtime variables are replaced by braced placeholders in capital letters (Table 6); all other characters, including typographical irregularities, are reproduced exactly as sent to the models and marked in footnotes rather than corrected. For each stage we show one canonical template in full; language variants are reported as verbatim line-level deltas. Superseded pilot prompts and auxiliary preprocessing prompts, which did not contribute to the released corpus, are omitted.[2]

Generation conditions.

- **Generation:** `gpt-3.5-turbo` and `gpt-4` for the English batches; `gemini-2.0-flash-lite` for the Russian and Kazakh batches. For `gemini-2.0-flash-lite`, default API decoding parameters (no temperature or top-p overrides set).
- **Judge:** `claude-haiku-4-5-20251001` (temperature 0, `max_tokens` 300) and `deepseek-chat` (temperature 0); identical prompt for both.
- **Audio/text detection:** all five decoders (Table 8) use greedy decoding with `max_new_tokens` 256.

[2]The dataset (https://huggingface.co/datasets/maristombayeva/lost-in-speech) includes the LLM-rewritten texts, type/severity and binary labels, source references, ASR transcripts and synthetic audio for the rewritten items in all three languages, the split manifests of both training runs, and encoder predictions. The dataset card documents record-level exceptions and access conditions.

Placeholder	Runtime value
{ARTICLE TEXT}	Source news article passed to the generation model.
{SEVERITY LEVEL DESCRIPTION}	Full mild/moderate/severe definition string (severity-definitions box).
{HALLUCINATION TYPE}	`contradiction` / `fabrication` / `context_inconsistency`.
{SEVERITY LEVEL}	`mild` / `moderate` / `severe`.
{SEVERITY LEVEL DEFINITION}	Judge-side one-sentence definition of the (type, level) pair.
{DIMENSION LIST} / {DIMENSION KEYS}	Type-dependent dimension names (list / quoted JSON keys).
{ORIGINAL ARTICLE TEXT}, {REWRITTEN ARTICLE TEXT}	Article pair for the judge (each truncated to 6,000 chars).
{INPUT TEXT}	Text passage under detection.
{DIGITS RULE}	`Remove all digits.` or `Keep digits.`

Table 6: Placeholders used in the prompt templates.

A.1 Hallucination Generation

Prompt for factual fabrication generation (canonical, English)

```
You are a professional news editor tasked with
generating factual fabrications at varying
levels. Here is the original news article:

Original news: {ARTICLE TEXT}

Follow this structure:
1. Extract the narrative, factual information,
entities, and contextual data from the
original news.
2. Create one version of the news:
   - {SEVERITY LEVEL DESCRIPTION}

3. Provide a report:
   - Factual accuracy
   - Credibility
   - Fluency
   - Coherence
   - Relevance
   - Satisfaction with fabrication level
   - Adequacy of type

   Evaluate each dimension as: Low, Very Low,
Medium, High, Very High.

4. Identify the fabricated sentences and
explain the deviations.

5. Make sure to provide the **full text of the
news with fabrications**, without hyperlinks:
   - {SEVERITY LEVEL DESCRIPTION} (full text):

Generate the answer strictly in English and
strictly following this format.
```

Instantiated once per article and severity level; the full news text is extracted by matching the "(full text)" header.

Prompt for factual contradiction generation (canonical, English)

```
You are a professional news editor tasked with
generating factual contradictions at varying
levels. Here is the original news article:

Original news: {ARTICLE TEXT}

Follow this structure:
1. Extract the narrative, factual information,
entities, and contextual data from the
original news.
2. Create one version of the news focusing on
factual contradictions:
    - {SEVERITY LEVEL DESCRIPTION}

3. Provide a report:
    - Factual accuracy
    - Credibility
    - Fluency
    - Coherence
    - Relevance
    - Satisfaction with contradiction level
    - Adequacy of type

    Evaluate each dimension as: Low, Very Low,
Medium, High, Very High.

4. Identify the sentences containing factual
contradictions and explain the deviations from
the original text and real-world facts.

5. Make sure to provide the **full text of the
news with factual contradictions**, without
hyperlinks:
    - {SEVERITY LEVEL DESCRIPTION} (full
text):

Generate the answer strictly in English and
strictly following this format.
```

Language-variant deltas: fabrication and contradiction (RU/KZ)

```
The RU/KZ variants differ from the canonical
templates ONLY in the following verbatim
lines.

[Step 1] "...from the original news" is
extended with the target language:
1. Extract the narrative, factual information,
entities, and contextual data from the
original news in **Russian**.
1. Extract the narrative, factual information,
entities, and contextual data from the
original news in **Kazakh**.

[Step 2] An additional bullet is appended
after {SEVERITY LEVEL DESCRIPTION}:
```

```
    - **Ensure the length of the generated
news text is similar to the original. Avoid
shortening or excessively extending the
article. Maintain paragraph structure and
detail level.**

[Step 3] Header and scale are localized
(dimension names keep trailing colons):
3. Provide a report evaluation in **Russian**:
    Evaluate each dimension as: Низкая, Очень
низкая, Средняя, Высокая, Очень высокая.
3. Provide a report evaluation in **Kazakh**:
KZ fabrication scale:
    Evaluate each dimension as: Өте төмен,
Төмен, Орташа, Жоғары, Өте жоғары
KZ contradiction scale:
    Evaluate each dimension as: Өте төмен,
Төмен, Орташа, Жоғары, Өте жоғары.

[Step 4] "...explain the deviations in
**Russian**." / "...real-world facts in
**Russian**." (same pattern with **Kazakh**)

[Step 5] Output-language requirement and
localized full-text markers:
5. Make sure to provide the **full text of
the news with fabrications** in **Russian**,
without hyperlinks:
    - {SEVERITY LEVEL DESCRIPTION} (полный
текст):
KZ fabrication marker:
    - {SEVERITY LEVEL DESCRIPTION} (толық
мәтін):
KZ contradiction marker:
    - {SEVERITY LEVEL DESCRIPTION} (толық
мәтін):

[Closing line]
Generate the answer strictly in English for
the instructions and strictly following
this format. The report evaluation and the
full text with fabrications should be in
**Russian**.
Generate the answer strictly in English for
the instructions and strictly following
this format. The report evaluation and the
full text with fabrications should be in
**Kazakh**.
Generate the answer strictly in English for
the instructions and strictly following this
format. The report evaluation and the full
text with factual contradictions should be in
**Russian**.
Generate the answer strictly in English for
the instructions and strictly following this
format. The report evaluation and the full
text with factual contradictions should be in
**Kazakh**.
```

Translation and transliteration note. The non-English strings appearing above are translated and transliterated as follows. Russian "Низкая, Очень низкая, Средняя, Высокая, Очень высокая" (*Nizkaya, Ochen' nizkaya, Srednyaya, Vysokaya, Ochen' vysokaya*) means "Low, Very Low, Medium, High, Very High". Kazakh "Өте төмен, Төмен, Орташа, Жоғары, Өте жоғары" (*Öte tömen, Tömen, Ortasha, Zhogary, Öte zhogary*) means "Very Low, Low, Medium, High, Very High". Russian "полный текст" (*polnyy tekst*) and Kazakh "толық мәтін" (*tolyq mätin*) both mean "full text".

Prompt for context inconsistency generation (canonical, English)

```
You are a professional news editor tasked
with generating a rewritten news article that
introduces **context inconsistency** at a
specified level. You will be provided with an
original article.

Your goal is to alter the logical or narrative
structure (entities, causality, timeline,
etc.) while keeping all **explicit numeric
facts** untouched.

Original news article:
{ARTICLE TEXT}

Your task:

1. Carefully extract and preserve all
**explicit numeric values** from the original
article. This includes:
   - Years and dates (e.g., 2023, July 15)
   - Quantities (e.g., 5,000 employees)
   - Money amounts (e.g., $1.2 billion)
   - Percentages (e.g., 13.7%)
   - Measurable data (e.g., 75 km, 3.6 GHz)

2. Create a **rewritten version** of
the article that demonstrates **context
inconsistency** as described below:
    - {SEVERITY LEVEL DESCRIPTION}

    Critical rules:
    - DO NOT add, remove, change, or
contradict ANY numeric values from the
original article. They must appear exactly
as in the original.
    - All inconsistencies must arise from
contradictions or distortions in logic,
narrative flow, cause-effect chains,
timeline, or facts - but NEVER from numerical
manipulation.
    - Output length must be within ±15% of the
original article length.

3. Provide a quality report evaluating the
rewritten article:
    - Internal Consistency
    - Logical Coherence
    - Fluency
    - Overall Coherence
```

```
    - Relevance to Original News
    - Satisfaction with Inconsistency Level
    - Adequacy of Inconsistency Type
    Rate each using: Very Low, Low, Medium,
High, Very High.

4. Identify and list all sentences
that introduce contradictions, context
misinterpretations, or logical flaws. Explain
why they are inconsistent with the original
article.

5. Output the full rewritten news article with
the specified inconsistency level:

{SEVERITY LEVEL DESCRIPTION} (full text):
```

Unlike the other two types, this template forbids numeric changes, adds a $\pm 15\%$ length constraint, and rates Internal Consistency and Logical Coherence instead of Factual accuracy and Credibility.

Language-variant deltas: context inconsistency (RU/KZ)

```
The RU/KZ variants differ from the canonical
template ONLY in the following verbatim lines.

[Intro] "...an original article in Russian." /
"...an original article in Kazakh."
Original news article (in Russian):
Original news article (in Kazakh):

[Step 1] Localized numeric examples:
    - Years and dates (e.g., 2023, 15 июля)
    - Quantities (e.g., 5 000 сотрудников)
    - Money amounts (e.g., $1.2 миллиарда)
    - Measurable data (e.g., 75 км, 3.6 ГГц)
    - Years and dates (e.g., 2023, 15 шілде)
    - Quantities (e.g., 5000 қызметкер)
    - Money amounts (e.g., $1,2 млрд)

[Step 2]
2. Create a **rewritten version** of the
article (in Russian) that demonstrates
**context inconsistency** as described below:
(same pattern with "(in Kazakh)")

[Step 3] Dimension names keep trailing colons;
localized scales:
    Rate each using: Очень низкий, Низкий,
Средний, Высокий, Очень высокий.
    Rate each using: Өте төмен, Төмен, Орташа,
Жоғары, Өте жоғары.

[Step 4]
4. Identify and list all sentences (in
Russian) that introduce contradictions,
context misinterpretations, or logical flaws.
Explain (in Russian) why they are inconsistent
with the original article.
(same pattern with "(in Kazakh)")

[Step 5]
5. Output the full rewritten news article
with the specified inconsistency level (in
Russian):
{SEVERITY LEVEL DESCRIPTION} (полный текст на
русском языке):
5. Output the full rewritten news article with
the specified inconsistency level (in Kazakh):
{SEVERITY LEVEL DESCRIPTION} (толық мәтіні
қазақ тілінде):
```

Translation and transliteration note. The additional non-English examples above are translated and transliterated as follows. Russian "15 июля" (*15 iyulya*), "5 000 сотрудников" (*5 000 sotrudnikov*), "$1.2 миллиарда" (*$1.2 milliarda*), and "75 км, 3.6 ГГц" (*75 km, 3.6 GGts*) mean "15 July", "5,000 employees", "$1.2 billion", and "75 km, 3.6 GHz", respectively. Kazakh "15 шілде" (*15 shilde*), "5000 қызметкер" (*5000 qyzmetker*), and "$1,2 млрд" (*$1,2 mlrd*) mean "15 July", "5,000 employees", and "$1.2 billion", respectively. Russian "полный текст на русском языке" (*polnyy tekst na russkom yazyke*) means "full text in Russian", and Kazakh "толық мәтіні қазақ тілінде" (*tolyq mätini qazaq tilinde*) means "full text in Kazakh". The localized rating scales use the same translations and transliterations as reported above.

Severity-level definitions inserted as {SEVERITY LEVEL DESCRIPTION}

```
Factual fabrication:
- Mild fabrication: Introduce small factual
deviations (e.g., slightly incorrect dates,
locations, or names) that are still plausible
and do not alter the overall message of the
article.
- Moderate fabrication: Introduce multiple
fictional or unverifiable statements that are
plausible but subtly distort the historical or
factual context of the article.
- Severe fabrication: Rewrite the article by
introducing clearly false or unverifiable
claims that significantly contradict
established facts or distort the original
narrative.


Factual fabrication - Moderate/Severe variants
used for the RU/KZ Gemini batches:
- Moderate fabrication: Introduce multiple
fictional or unverifiable statements that are
plausible and subtly distort the historical
or factual context of the article. Ensure
the narrative remains within the realm of
possibility.
- Severe fabrication: Rewrite the article by
introducing significant factual errors and
unverifiable claims that substantially alter
the narrative. However, maintain a degree
of realism and avoid overtly fantastical or
scientifically impossible elements. Focus on
```

```
plausible alternative scenarios or large-scale
misinformation.

Factual contradiction (all languages):
- Mild factual contradiction: Introduce minor
factual errors involving entities or relations
that directly contradict easily verifiable
information but maintain the overall narrative
structure.
- Moderate factual contradiction: Introduce
several factual errors involving key entities
and their relationships, significantly
distorting the factual basis of the news
while maintaining a degree of plausibility in
the narrative.
- Severe factual contradiction: Completely
rewrite significant portions of the news
with demonstrably false and contradictory
information regarding core entities, events,
and their relationships, resulting in a highly
inaccurate account.

Context inconsistency (all languages):
- Mild context inconsistency: Introduce
minor contradictions that deviate from
specific details or assumptions in the
original article, while largely preserving
its narrative flow.
- Moderate context inconsistency: Introduce
several contradictions that directly oppose
key elements or implications of the original
article, potentially creating some context
inconsistencies.
- Severe context inconsistency: Generate a
version of the article that fundamentally
disregards and contradicts the original
article's internal context, introducing major
logical flaws and factual distortions.
```

The fabrication definitions were revised between the English batch and the later Russian/Kazakh batches; both wordings contributed articles to the corpus and are therefore reported.

A.2 LLM-as-a-Judge Quality Evaluation

Prompt for LLM-as-a-judge quality evaluation (Claude and DeepSeek)

```
You are a professional news-quality evaluator
(judge only; do NOT rewrite the article).
You are given the ORIGINAL article, the
intended hallucination TYPE and SEVERITY,
and a REWRITTEN article.
Rate the REWRITTEN article on each dimension
using EXACTLY one of:
Very Low, Low, Medium, High, Very High.

TYPE: {HALLUCINATION TYPE}
SEVERITY: {SEVERITY LEVEL} - {SEVERITY LEVEL
DEFINITION}

Dimensions:
{DIMENSION LIST}

Return ONLY a JSON object mapping each
dimension name to its rating, using these
exact keys:
{{DIMENSION KEYS}}

ORIGINAL:
{ORIGINAL ARTICLE TEXT}

REWRITTEN:
{REWRITTEN ARTICLE TEXT}
```

Type-dependent dimension sets inserted as {DIMENSION LIST} / {DIMENSION KEYS}

```
contradiction: Factual accuracy (FA),
Credibility (Cr), Fluency (Fl), Coherence
(Co), Relevance (Re), Satisfaction with
contradiction level (Sa), Adequacy of type
(AT)

fabrication: Factual accuracy (FA),
Credibility (Cr), Fluency (Fl), Coherence
(Co), Relevance (Re), Satisfaction with
fabrication level (Sa), Adequacy of type
(AT)

context_inconsistency: Internal Consistency
(IC), Logical Coherence (LC), Fluency (Fl),
Overall Coherence (Co), Relevance to Original
News (Re), Satisfaction with Inconsistency
Level (Sa), Adequacy of Inconsistency Type
(AT)
```

Judge-side severity definitions inserted as {SEVERITY LEVEL DEFINITION}

```
contradiction:
- mild: Mild factual contradiction: minor
factual errors involving entities or relations
that directly contradict easily verifiable
information but maintain the overall narrative
structure.
- moderate: Moderate factual contradiction:
several factual errors involving key
entities and relationships, significantly
distorting the factual basis while keeping
some plausibility.
- severe: Severe factual contradiction:
rewrite significant portions with demonstrably
false, contradictory information about core
entities, events, and relationships.

fabrication:
- mild: Mild fabrication: small plausible
factual deviations (e.g., slightly incorrect
dates, locations, names) that do not alter the
overall message.
- moderate: Moderate fabrication: multiple
fictional or unverifiable but plausible
statements that subtly distort the factual
context.
- severe: Severe fabrication: clearly false,
unverifiable, or fantastical claims that
significantly contradict real-world knowledge.
```

```
context_inconsistency:
- mild: Mild context inconsistency: subtle
shifts in emphasis or context that slightly
distort meaning without explicit factual
errors.
- moderate: Moderate context inconsistency:
noticeable distortions of context/emphasis
that change interpretation.
- severe: Severe context inconsistency: strong
contextual distortions that substantially
change the meaning of the article.
```

Both judges belong to model families disjoint from the generators. Verbal ratings are mapped to integers 1–5 during parsing (including Kazakh scale variants); the JSON object is extracted with a regular expression, with up to three retries on parse failure.

A.3 Zero-Shot Detection (All Decoders, Audio and Text)

System prompt for Qwen2.5-Omni

```
You are Qwen, a virtual human developed by
the Qwen Team, Alibaba Group, capable of
perceiving auditory and visual inputs, as well
as generating text and speech.
```

All five decoders receive the same user-turn prompts below. The system prompt above is the default system prompt of Qwen2.5-Omni; the other decoders use their default system prompt.

Prompt for direct zero-shot detection, binary task (canonical, audio input)

```
You are an expert assistant specialized
in analyzing audio content and detecting
hallucinations.

Your task is to determine whether there are
any hallucinations in the audio content.

A hallucination is any statement that contains
factual contradictions, fabricated details,
or contextual inconsistencies relative to the
actual audio content.

Given the audio content, determine:
Is there a hallucination? (yes/no)

Output format: {"binary": "yes/no"}
```

Type and severity task variants (verbatim components)

```
The type and severity prompts follow the same
structure as the binary prompt, with the
following verbatim substitutions.

TYPE task - role, task statement, definitions,
and closing:
You are an expert assistant specialized in
analyzing audio content and classifying
hallucination types.
Your task is to identify the type of
hallucination present in the audio content.

Hallucination Types:
- Factual Contradiction: Statements that
directly conflict with known facts or
information provided in the audio content.
- Factual Fabrication: Insertion of fabricated
yet plausible-sounding details not grounded in
the audio content.
- Contextual Inconsistency: Subtle alterations
that distort the meaning, emphasis, or context
of the audio content without introducing
explicit factual errors.

Given the audio content, classify the
hallucination type:
What type of hallucination is present?
(factual_contradiction/factual_fabrication/
contextual_inconsistency/none)

Output format: {"type":
"factual_contradiction/factual_fabrication/
contextual_inconsistency/none"}

SEVERITY task - role, task statement,
definitions, and closing:
You are an expert assistant specialized
in analyzing audio content and assessing
hallucination severity.
Your task is to assess the severity level
of any hallucination present in the audio
content.

Severity Levels:
- Mild: Subtle distortions or minor deviations
that preserve the main narrative and
plausibility of the audio content.
- Moderate: Noticeable inconsistencies or
factual alterations that affect key details or
context while maintaining partial alignment
with the audio content.
- Severe: Major contradictions, fabrications,
or contextual breakdowns that substantially
misrepresent or conflict with the audio
content's facts or intent.

Given the audio content, classify the
hallucination severity:
What is the severity level? (mild/moderate/
severe/none)

Output format: {"degree": "mild/moderate/
severe/none"}
```

Chain-of-thought variants (verbatim replaced closing lines)

```
In the CoT condition the direct closing lines
are replaced as follows (Output format lines
unchanged).

BINARY:
Given the audio content, think step-by-step
about whether hallucinations are present:
```

```
Then provide your final answer:
Is there a hallucination? (yes/no)

TYPE:
Given the audio content, think step-by-step
about what type of hallucination may be
present:

Then provide your final classification:
What type of hallucination is present?
(factual_contradiction/factual_fabrication/
contextual_inconsistency/none)

SEVERITY:
Given the audio content, think step-by-step
about the severity of any hallucination:

Then provide your final assessment:
What is the severity level? (mild/moderate/
severe/none)
```

Each task is a separate call under both conditions. Text-input variants replace "audio content" with "text content"; in the text condition the passage is appended as "`Text to analyze: {INPUT TEXT}`", while in the audio condition the audio file is attached to the user turn. Predictions are parsed from the JSON fragment of the response.

A.4 ASR Transcript Cleaning for WER/CER

Prompt for ASR transcript cleaning for WER/CER (canonical, single item)

```
Clean ASR transcript for WER/CER evaluation.

Rules:
- Do NOT add, paraphrase, or translate.
- Lowercase.
- Remove punctuation, quotes, brackets, noise
tokens ([music], <unk>), timestamps (e.g.,
00:01:23).
- Replace hyphens with spaces.
- Collapse multiple spaces.
- {DIGITS RULE}

Return ONLY cleaned text.
```

Batched (JSONL) cleaning variant (verbatim deltas)

```
The batched variant differs in the following
verbatim lines:

You clean ASR transcripts for fair WER/CER
evaluation.
- Do NOT add new content, do NOT paraphrase,
do NOT translate.
- Remove punctuation, noise tokens and
timestamps (e.g., 00:01:23).

OUTPUT FORMAT (very important):
Return ONLY JSONL: one JSON object per line,
no surrounding array, no markdown.
Each line must be:
{"idx": <int>, "cleaned": "<string>"}
```

Cleaning normalizes both reference and hypothesis strings before WER/CER computation; batches are split recursively on JSON parse failures.

B Training Details

Encoders are fine-tuned for the binary task with effective batch size 16 (batch 4, gradient accumulation 4), learning rate 2e-5, 3 epochs, maximum sequence length 512, fp16 with gradient checkpointing, and early stopping on development macro-F1 (patience 1); seed 42. For mDeBERTa-v3, which is unstable under fp16, we instead use fp32, eager attention, learning rate 1e-5 with 10% linear warmup, and physical batch 8 with gradient accumulation 2 (same effective batch 16). Data are split 80/10/10 (train/dev/test), stratified by language and label. The encoders evaluated on the real-world split (Table 4) are trained in a dedicated run in which the 580 truthful negatives (290 per language, drawn from the ru/kz non-hallucinated class of the corpus) are removed from the train/dev pool before splitting and appended to the synthetic test split, ensuring that no model evaluated on real-world data was trained on its truthful items. Table 2 reports an independent run of the same configuration on the full corpus split; cross-table differences in the binary figures therefore combine training-pool, test-composition, and run-to-run effects, and within-table comparisons always use a single run. The split files of both runs are included in the released dataset. For the transcript (T) columns of Table 4, both encoders are fine-tuned on the ASR transcripts of the synthetic corpus using the identical train/dev/test split (by article id) as the original-text models.

C Models

Table 8 lists all detection, generation, and speech models used in the study, with their modality support. Table 7 details the per-language ASR selection discussed in Section 3.3.

D Full LLM-Based Quality Evaluation

Table 10 reports the complete LLM-based quality evaluation by hallucination type, severity, and language, discussed in Section 3.4. Table 9 reports per-dimension agreement between the two judges.

Table 7: Per-language ASR error (WER/CER, %, corpus level) on identical synthesized audio: a single multilingual model (Whisper-large-v3) vs. our per-language selection. Whisper-large-v3 fails on low-resource Kazakh, while a fine-tuned wav2vec2 model nearly halves the Kazakh error rate; we therefore use Whisper-large-v3 for English/Russian and fine-tuned wav2vec2 for Kazakh.

ASR system	**EN** WER/CER	**RU** WER/CER	**KZ** WER/CER
Whisper-large-v3	**7.30/2.74**	**9.08/4.74**	64.27/32.03
wav2vec2 (fine-tuned)	–	–	**34.04/16.90**
Used in benchmark	Whisper	Whisper	wav2vec2

E Real-World Split Details

The 290 fake items were downloaded from the public factcheck.kz archive in October 2024 (225 natively Russian, 65 natively Kazakh, after removal of 15 exact duplicates from the initial Russian collection). Each item was machine-translated into the other language with Google Translate; a fluent bilingual author manually reviewed all 290 translations and judged all of them adequate, with no post-editing required. Truthful negatives (290 per language) are original human-written articles of the synthetic subset, selected via greedy TF-IDF cosine matching to the fakes; they are excluded from the train/dev splits of all models evaluated on the real-world data (Appendix B). The faithful-rewrite condition of Table 5 rewrites the same 290 Russian truthful negatives with Gemini 2.0 Flash-Lite under a hallucination-free instruction (preserve all facts, names, numbers, dates, and claims; introduce no unsupported information; keep approximately the same length).

Table 8: Models used in the study. P = Purpose (○ Detection, □ Text generation, △ Audio/TTS/ASR). Type: Enc = Encoder, Dec = Decoder. A = Audio support, T = Text support. **Lang** = languages with officially documented support, per the model cards or technical reports; for audio decoders the figure before the slash refers to text and the figure after it to speech input (n/r = not reported by the developers). TTS/ASR rows give supported languages, not the single language each system is used for here (Section 3.3). ‡Gemma-3n-E4B has 8B raw parameters and runs with a memory footprint comparable to a 4B model.

Model	**P**	**Type**	**Lang**	**Params**	**Year**	**A**	**T**
Detection Models							
XLM-R (base)	○	Enc	100	279M	2019	✗	✓
XLM-R (large)	○	Enc	100	561M	2019	✗	✓
mDeBERTa	○	Enc	100	279M	2021	✗	✓
ReMBERT	○	Enc	110	576M	2021	✗	✓
Qwen2.5-Omni	○	Dec	29 / n/r	3B	2025	✓	✓
Qwen2-Audio	○	Dec	8 / n/r	7B	2024	✓	✓
Gemma 3n	○	Dec	140 / 35	4B‡	2025	✓	✓
LFM2-Audio	○	Dec	en / en	1.5B	2025	✓	✓
Step-Audio-2-mini	○	Dec	zh,en / zh,en	8B	2025	✓	✓
Generation Models							
GPT-3.5-turbo	□	Dec	100+	–	2023	✗	✓
GPT-4	□	Dec	100+	–	2023	✗	✓
Gemini 2.0 Flash-Lite	□	Dec	100+	–	2025	✗	✓
Coqui XTTS-v2	△	TTS	17 (en used)	–	2023	✓	✗
Silero TTS	△	TTS	multi (ru used)	–	2021	✓	✗
Kazakh TTS	△	TTS	kz	50M	2021	✓	✗
Whisper-large-v3	△	ASR	99	1550M	2023	✓	✓
wav2vec2 (fine-tuned)	△	ASR	kz	–	2024	✓	✓

Table 9: Cross-judge agreement on the stratified sample between the two external judges, Claude Haiku 4.5 and DeepSeek Chat, per quality dimension. κ: quadratic-weighted Cohen's κ; ρ: Spearman correlation; W1: agreement within one point. Ordered by κ.

Dimension	n	κ	ρ	**W1**
Credibility (Cr)	144	0.897	0.923	1.00
Factual Accuracy (FA)	144	0.875	0.910	1.00
Relevance (Re)	216	0.676	0.721	0.92
Satisfaction (Sa)	216	0.600	0.674	0.97
Coherence (Co)	216	0.591	0.610	0.92
Adequacy of Type (AT)	216	0.511	0.497	0.97
Fluency (Fl)	216	0.311	0.390	0.92
Logical Coherence (LC)	72	0.144	0.258	0.78
Internal Consistency (IC)	72	0.110	0.175	0.71

Table 10: LLM-as-a-judge quality evaluation by hallucination type, severity, and language, computed by the independent judge (claude-haiku-4-5, temperature 0) on the stratified validation sample of Section 3.4 (216 items; n=8 per cell). Mi = Mild, Mo = Moderate, Se = Severe. Scores are on a 1 (low) to 5 (high) scale. FA and Cr coincided on every fabrication/contradiction item in this sample and are reported separately for completeness.

Type	Q	EN			KZ			RU		
		Mi	**Mo**	**Se**	**Mi**	**Mo**	**Se**	**Mi**	**Mo**	**Se**
	AT	2.88	4.25	5.00	4.00	4.25	4.88	3.88	4.50	5.00
	Co	1.75	2.00	1.50	2.12	2.25	1.88	1.88	2.50	1.25
	Fl	2.88	3.00	2.88	3.38	3.38	3.12	3.25	3.38	3.00
$\otimes$	IC	2.00	2.00	2.25	2.25	2.25	1.88	2.00	2.88	1.62
	LC	1.62	2.00	1.50	2.12	2.25	1.88	1.75	2.50	1.25
	Re	2.00	2.00	1.75	2.12	2.00	1.62	2.25	2.00	1.00
	Sa	3.88	4.25	5.00	4.00	4.25	4.88	4.00	4.50	5.00
	AT	4.00	4.12	5.00	2.62	4.12	5.00	4.25	4.25	5.00
	Co	4.00	3.75	2.38	4.25	3.88	3.00	4.38	4.00	3.38
	Cr	2.00	2.00	1.00	1.62	1.88	1.00	2.00	2.00	1.00
$\diamond$	FA	2.00	2.00	1.00	1.62	1.88	1.00	2.00	2.00	1.00
	Fl	4.00	3.88	2.88	4.25	4.00	3.75	4.38	4.00	4.00
	Re	4.00	3.38	2.25	2.12	3.50	2.62	4.25	3.25	2.62
	Sa	4.00	4.00	5.00	2.38	4.00	5.00	4.00	4.00	5.00
	AT	4.25	4.88	5.00	4.25	4.88	5.00	4.75	5.00	5.00
	Co	3.38	2.50	2.12	3.88	3.88	3.25	3.38	3.12	3.38
	Cr	1.50	1.12	1.00	1.75	1.12	1.00	1.25	1.00	1.00
$\neq$	FA	1.50	1.12	1.00	1.75	1.12	1.00	1.25	1.00	1.00
	Fl	3.62	3.12	2.62	4.00	4.00	3.75	3.75	3.75	4.00
	Re	3.50	2.62	2.12	3.75	3.25	2.38	3.38	2.75	2.88
	Sa	4.38	4.88	5.00	4.25	4.88	5.00	4.75	5.00	5.00

Legend: $\otimes$ = Context Inconsistency, $\diamond$ = Fabrication, $\neq$ = Contradiction. AT = Adequacy of Type, Fl = Fluency, IC = Internal Consistency, LC = Logical Coherence, Re = Relevance, Co = Coherence, Cr = Credibility, FA = Factual Accuracy, Sa = Satisfaction.